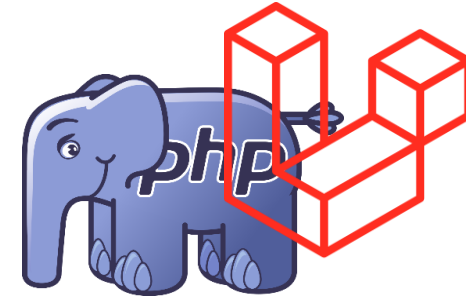




Getting Started with Laravel

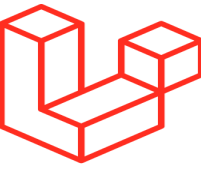
Prepared by

Oum Saokosal

Master's Degree in Information Systems,
South Korea, '2010

Tel/Telegram: **012 252 752**

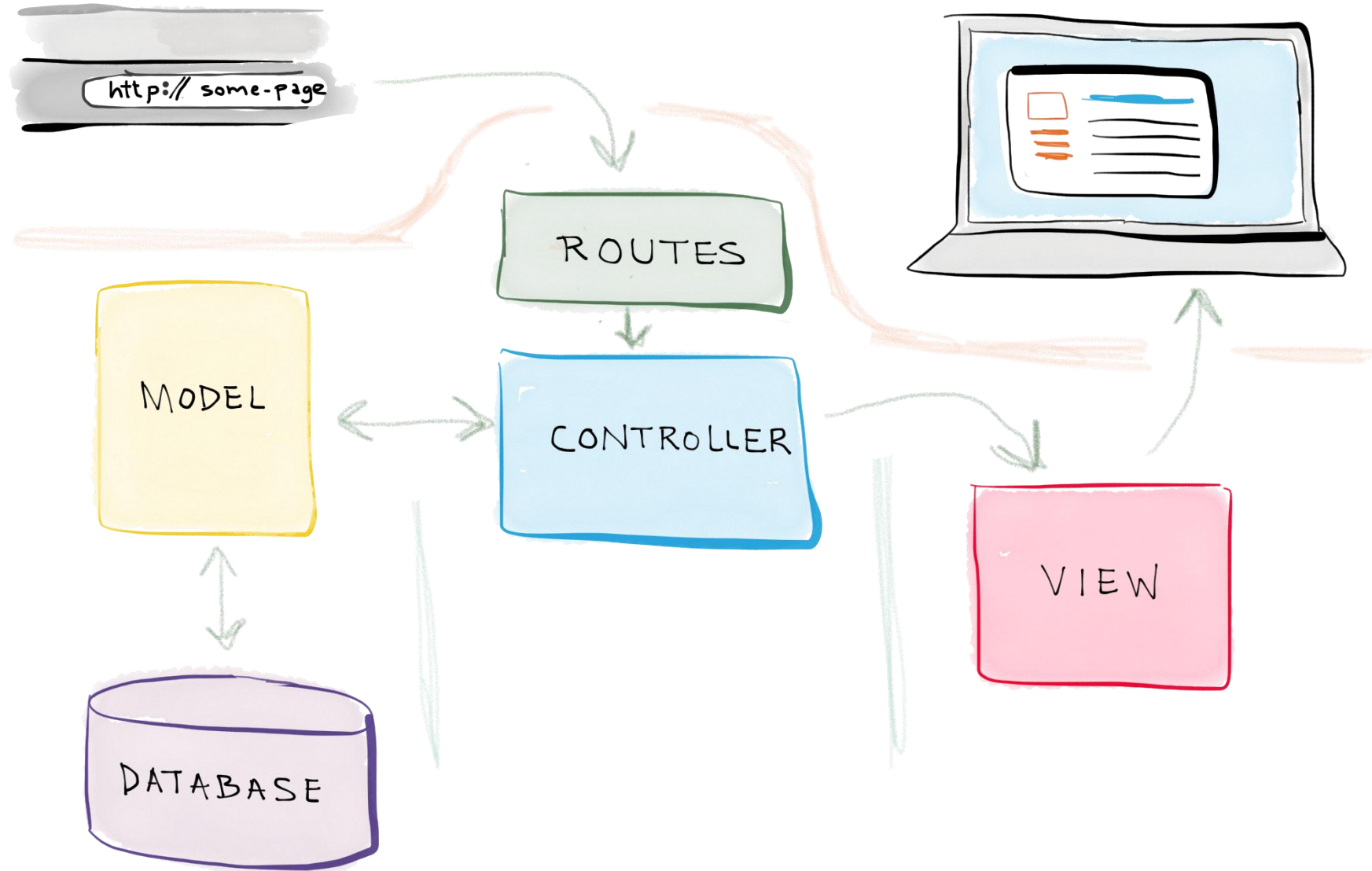
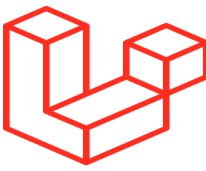
Laravel

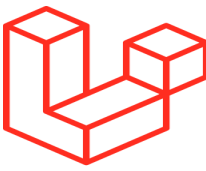


- Laravel is a free and open-source PHP-based web framework for building web applications. It was intended for the development of web applications following the model–view–controller (MVC) architectural pattern.
 - Created by **Taylor Otwell**.



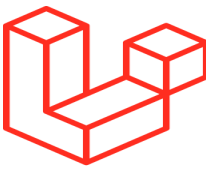
MVC and Route





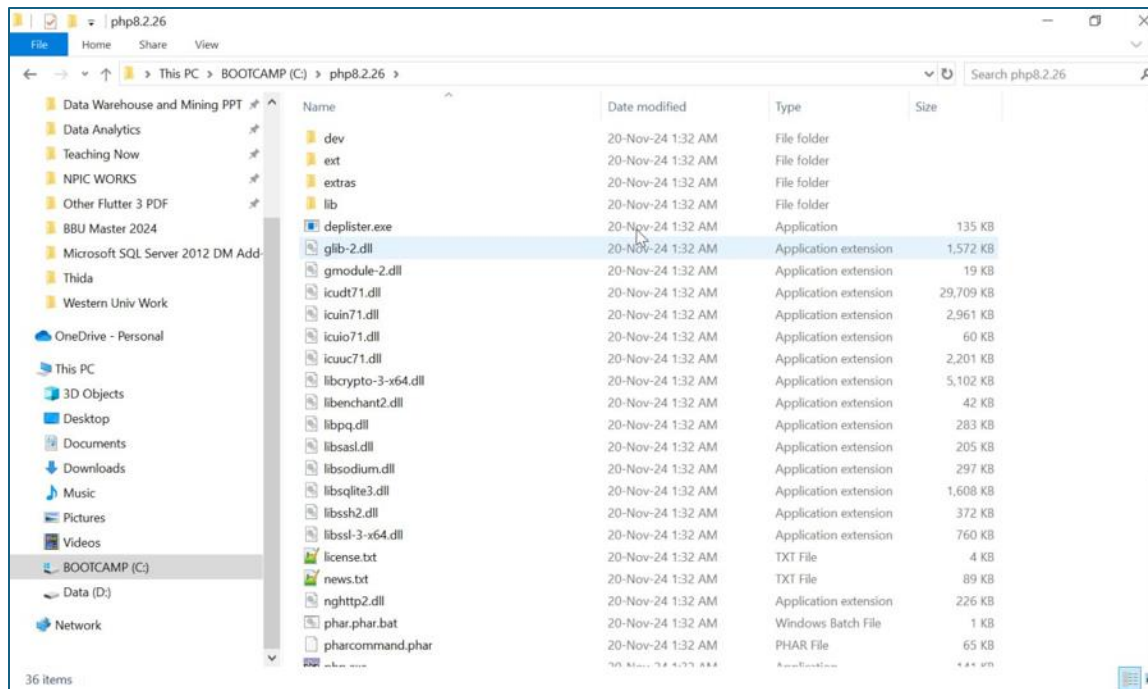
Why Use Laravel?

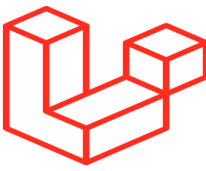
- **Elegant Syntax** – Laravel provides a clean and expressive syntax for developers.
- **MVC Architecture** – Separates concerns between Model, View, and Controller.
- **Built-in Features** – Authentication, routing, and database migration are built-in.
- **Eloquent ORM** – Simplifies database interactions with object-relational mapping (ORM).
- **Blade Templating** – Provides an efficient templating engine for views.
- **Security** – Features such as CSRF protection, hashing, and authentication mechanisms.



Config Laravel – PHP

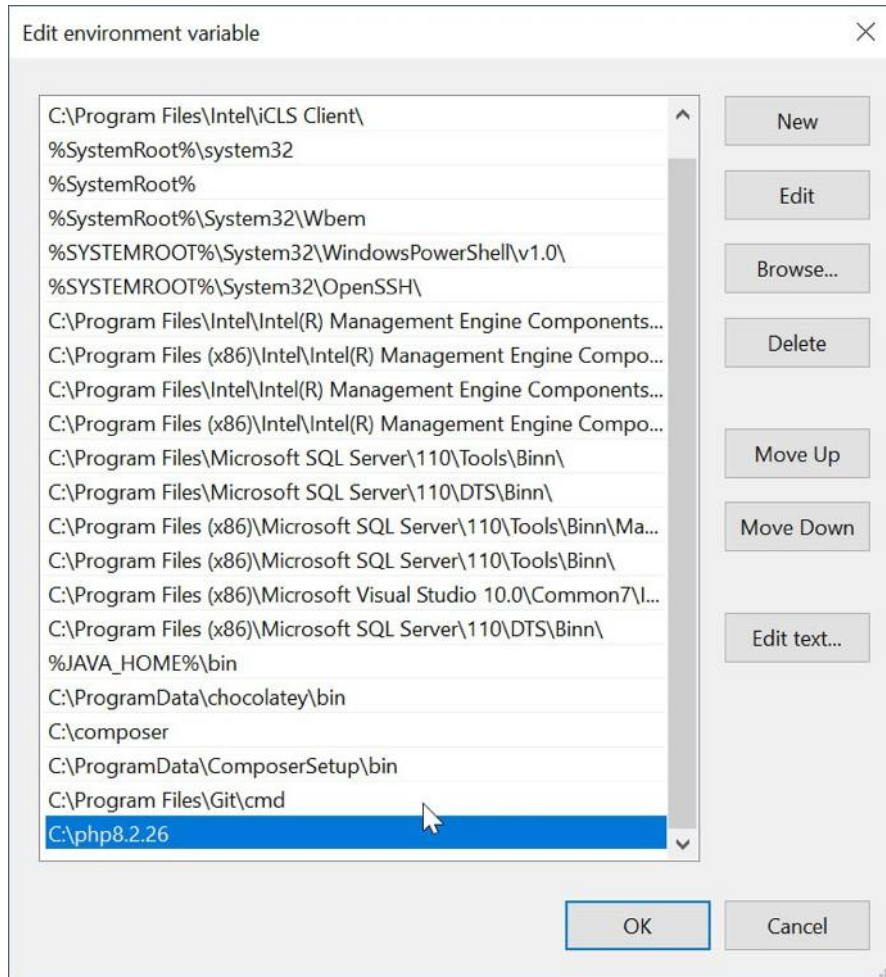
- First, you need to download PHP:
 - <https://www.php.net/downloads.php>
 - You may download the latest version
- Extract the zip file into drive C, e.g. C:\php8.2.28

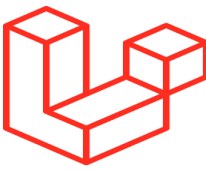




Config Laravel – PHP (cont.)

- Set PHP path in Environment Variable:





Config Laravel – PHP (cont.)

- You may need to download VC++ from <https://learn.microsoft.com/en-us/cpp/windows/latest-supported-vc-redist?view=msvc-170>

Version

Visual Studio 2022

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Redistribute Visual C++ ActiveX Controls

Redistribute the MFC Library

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redist, download the redist you're interested in using one of the following links. Then, look at its properties using Windows File Explorer. In the **Details** pane, the **File version** contains the version of the redist.

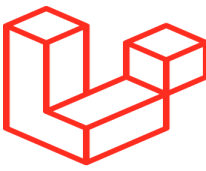
Latest Microsoft Visual C++ Redistributable version

The latest version is **v14.44.35208.0**

Use the following links to download this version for each supported architecture:

Expand table

Architecture	Link	Notes
ARM64	https://aka.ms/vs/17/release/vc_redist.arm64.exe	Permalink for latest supported ARM64 version
x86	https://aka.ms/vs/17/release/vc_redist.x86.exe	Permalink for latest supported x86 version
x64	https://aka.ms/vs/17/release/vc_redist.x64.exe	Permalink for latest supported x64 version. The X64 Redistributable package contains both ARM64 and X64 binaries. This package makes it easy to install required Visual C++ ARM64 binaries when

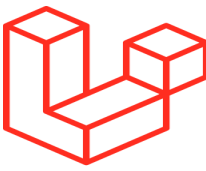


Config Laravel – PHP (cont.)

- Check PHP version and location:

```
php -v  
where php
```

```
Command Prompt  
Microsoft Windows [Version 10.0.19045.4651]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\kosal>php -v  
PHP 8.2.26 (cli) (built: Nov 19 2024 18:15:04) (NTS Visual C++ 2019 x64)  
Copyright (c) The PHP Group  
Zend Engine v4.2.26, Copyright (c) Zend Technologies  
  
C:\Users\kosal>where php  
C:\php8.2.26\php.exe  
  
C:\Users\kosal>
```

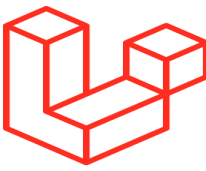
Config Laravel – Composer

- First you need to download **composer**.
 - *Composer is an application-level dependency manager for the PHP programming language.*
 - <https://getcomposer.org/download/>
- After the installation, you can check the version:

```
composer --version
```



PostgreSQL

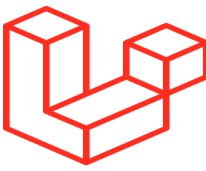


- Make sure you download [PostgreSQL](#)
- During the setup, password: 123456789 and port: 5432

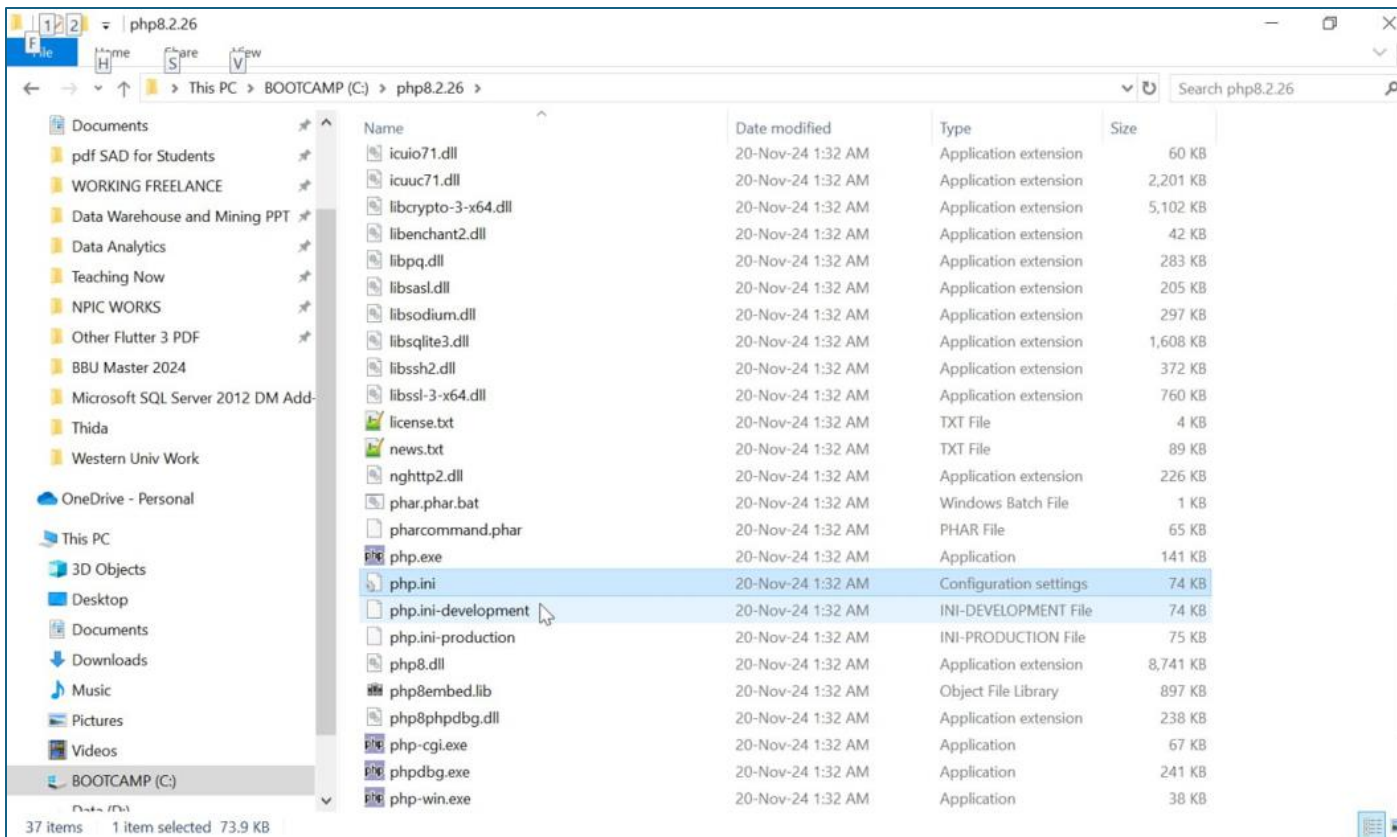
The screenshot shows the 'Password' step of the PostgreSQL setup. The window title is 'Setup'. It contains a green arrow icon pointing right. The text reads: 'Please provide a password for the database superuser (postgres)'. There are two input fields: 'Password' and 'Retype password', both containing seven asterisks. At the bottom, there are three buttons: '< Back', 'Next >', and 'Cancel'. The 'InstallBuilder' logo is in the bottom left corner.

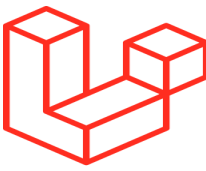
The screenshot shows the 'PostgreSQL Port' step of the PostgreSQL setup. The window title is 'Setup'. It contains a plus icon. The text reads: 'The installer will initialize the PostgreSQL database server with this port (leave as-is if you are unsure what to enter)'. There is an input field for 'PostgreSQL Server Port' with the value '5432' entered. At the bottom, there are three buttons: '< Back', 'Next >', and 'Cancel'. The 'InstallBuilder' logo is in the bottom left corner.

Config PHP ini



- Open the PHP folder
- Copy and paste **php.ini-development** then rename from to **php.ini**:

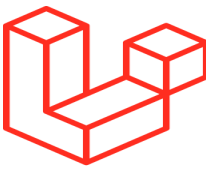




Config PHP ini (cont.)

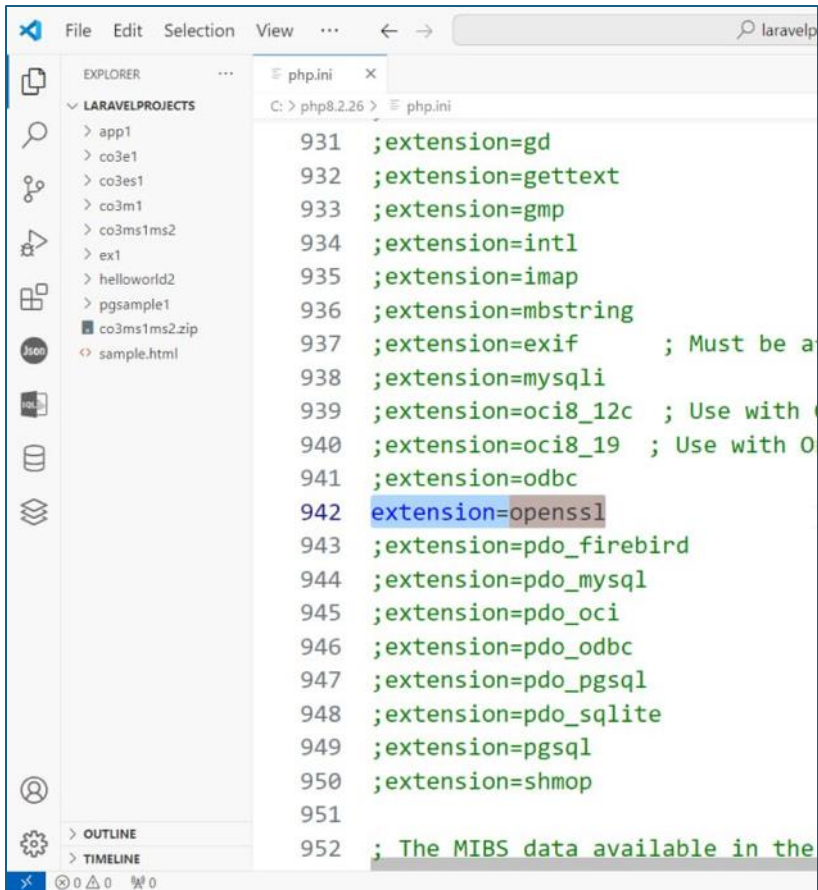
- Open the **php.ini** in VS Code or Notepad.
- Find **extension_dir = "ext"** and uncomment it

```
756 ; https://php.net/doc-root
757 doc_root =
758
759 ; The directory under which PHP opens the script using ~/username used only
760 ; if nonempty.
761 ; https://php.net/user-dir
762 user_dir =
763
764 ; Directory in which the loadable extensions (modules) reside.
765 ; https://php.net/extension-dir
766 ;extension_dir = "./"
767 ; On windows:
768 extension_dir = "ext"
769
770 ; Directory where the temporary files should be placed.
771 ; Defaults to the system default (see sys_get_temp_dir)
772 ;sys_temp_dir = "/tmp"
773
774 ; Whether or not to enable the dl() function. The dl() function does NOT wo
775 ; properly in multithreaded servers, such as IIS or Zeus, and is automatical
776 ; disabled on them.
777 ; https://php.net/enable-dl
```

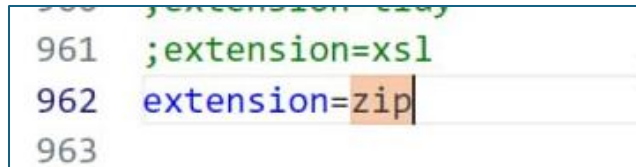


Config PHP ini (cont.)

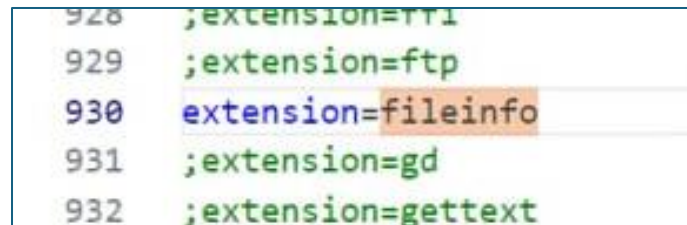
- Find **openssl**, **zip**, and **fileinfo**, and then uncomment them:



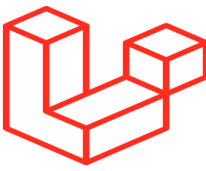
```
931 ;extension=gd
932 ;extension=gettext
933 ;extension=gmp
934 ;extension=intl
935 ;extension=imap
936 ;extension=mbstring
937 ;extension=exif      ; Must be a
938 ;extension=mysqli
939 ;extension=oci8_12c  ; Use with
940 ;extension=oci8_19  ; Use with O
941 ;extension=odbc
942 extension=openssl
943 ;extension=pdo_firebird
944 ;extension=pdo_mysql
945 ;extension=pdo_oci
946 ;extension=pdo_odbc
947 ;extension=pdo_pgsql
948 ;extension=pdo_sqlite
949 ;extension=pgsql
950 ;extension=shmop
951
952 ; The MIBS data available in the
```



```
961 ;extension=xsl
962 extension=zip
963
```



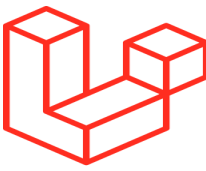
```
928 ;extension=ffi
929 ;extension=ftp
930 extension=fileinfo
931 ;extension=gd
932 ;extension=gettext
```



Config PHP ini (cont.)

- Because we're gonna use **PostgreSQL**, then we need to add **pgsql**

```
921 ;extension=ftp
922 ;extension=fileinfo
923 ;extension=gd
924 ;extension=gettext
925 ;extension=gmp
926 ;extension=intl
927 ;extension=ldap
928 extension=mbstring
929 ;extension=exif      ; Must be after mbstring as it depends on it
930 ;extension=mysqli
931 ;extension=odbc
932 extension=openssl
933 ;extension=pdo_firebird
934 ;extension=pdo_mysql
935 ;extension=pdo_odbc
936 extension=pdo_pgsql
937 ;extension=pdo_sqlite
938 extension=pgsql
939 ;extension=shmop
940
941 ; The MIBS data available in the PHP distribution must be installed.
942 ; See https://www.php.net/manual/en/snmp.installation.php
943 ;extension=snmp
```

Config PHP ini (cont.)

- After changed, then save the file.
- You need to run this command to reload the ini file:

```
php --ini
```

The screenshot shows a VS Code terminal window with the following content:

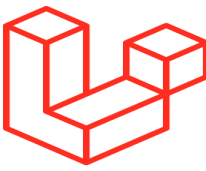
```
PROBLEMS  OUTPUT  TERMINAL  PORTS  DEBUG CONSOLE
```

Press Ctrl+C to stop the server

```
2024-11-30 09:43:24 / .....
2024-11-30 09:43:31 / .....
PS C:\laravelprojects\com3m1app1> php --ini ←
Configuration File (php.ini) Path:
Loaded Configuration File:      C:\php8.2.26\php.ini
Scan for additional .ini files in: (none)
Additional .ini files parsed:   (none)
PS C:\laravelprojects\com3m1app1> |
```

At the bottom of the terminal, there is a status bar that reads: "indexing laravel/framework:11.34.2 for dynamic references: 1170/1414 files".

PostgreSQL



- Make sure you download [PostgreSQL](#)
- During the setup, password: 123456789 and port: 5432

Setup

Password

Please provide a password for the database superuser (postgres).

Password

Retype password

InstallBuilder

< Back Next > Cancel

This screenshot shows the 'Password' step of the PostgreSQL setup. It prompts the user to provide a password for the database superuser (postgres). There are two input fields: 'Password' and 'Retype password', both containing masked text (*****). The window has a title bar with 'Setup' and standard minimize, maximize, and close buttons. A green arrow icon is visible in the top right corner. At the bottom, there are buttons for '< Back', 'Next >', and 'Cancel', along with the 'InstallBuilder' logo.

Setup

PostgreSQL Port

The installer will initialize the PostgreSQL database server with this port (leave as-is if you are unsure what to enter)

PostgreSQL Server Port

InstallBuilder

< Back Next > Cancel

This screenshot shows the 'PostgreSQL Port' step of the PostgreSQL setup. It informs the user that the installer will initialize the PostgreSQL database server with the specified port. The 'PostgreSQL Server Port' is set to '5432'. The window has a title bar with 'Setup' and standard minimize, maximize, and close buttons. At the bottom, there are buttons for '< Back', 'Next >', and 'Cancel', along with the 'InstallBuilder' logo.

Create a Laravel project

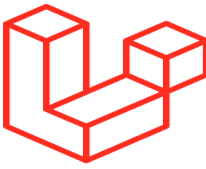
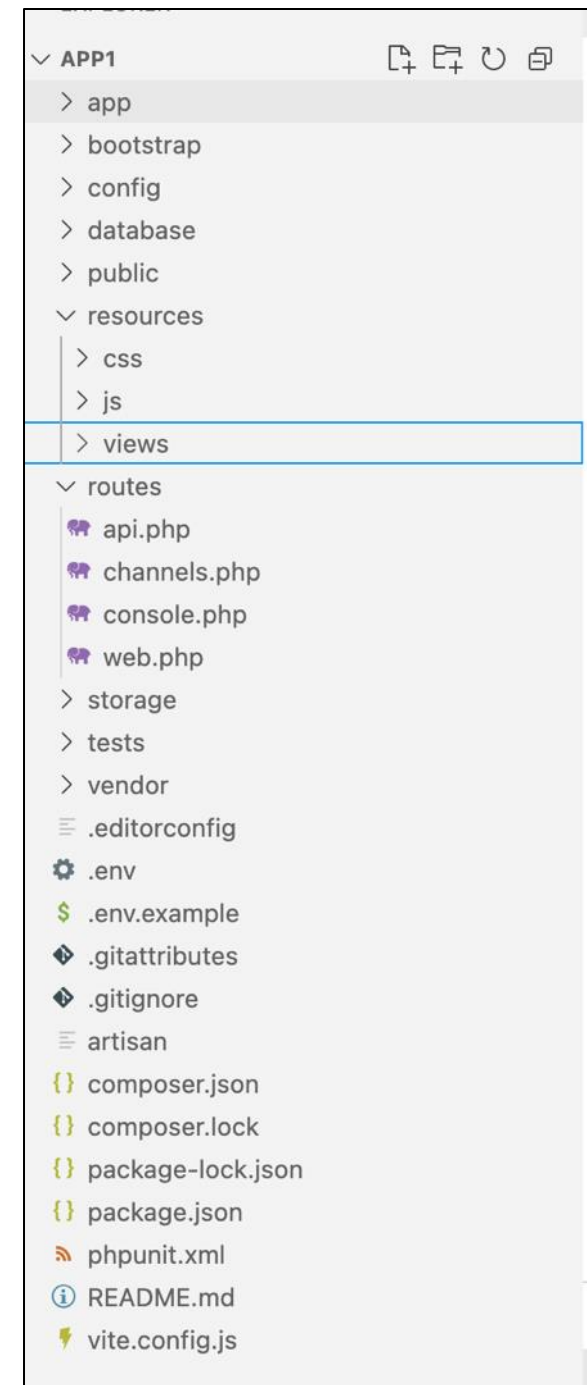
Create a laravel project:

```
composer create-project --prefer-dist laravel/laravel app1
```

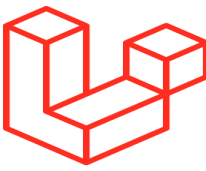
```
cd app1
```

```
code .
```

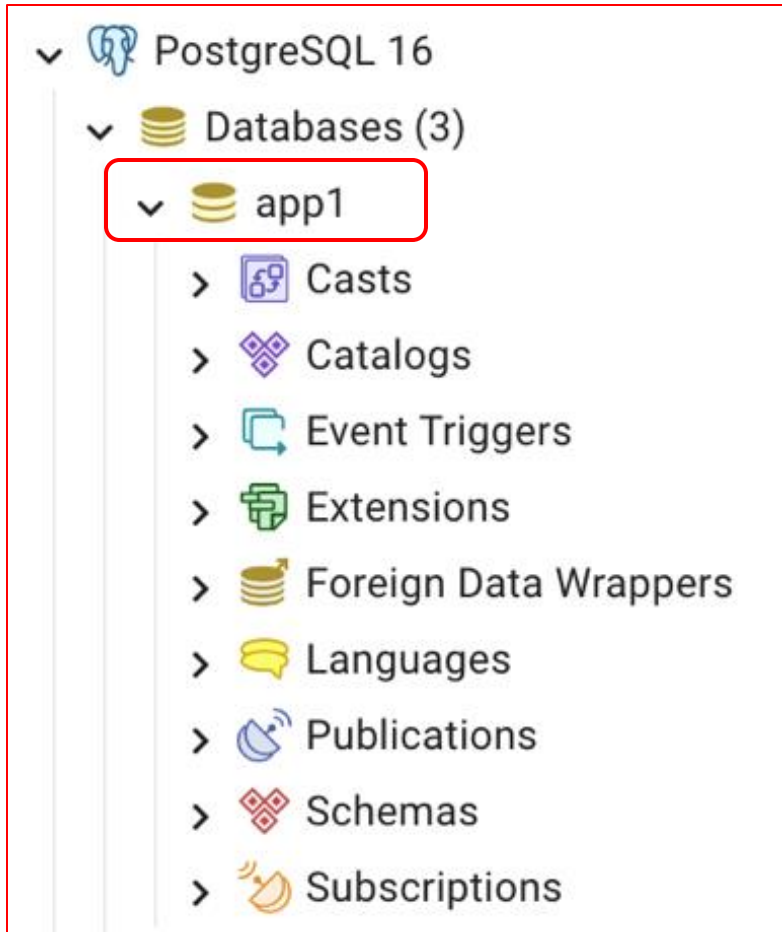
```
php artisan serve
```

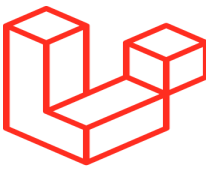


Create a database in Postgres



- Create a database and name it: **app1**



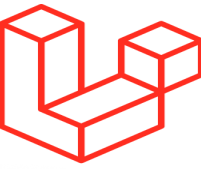


Config .ENV

- Config `.env` file, port: `5432` and Password: `123456789`

```
DB_CONNECTION=pgsql
DB_HOST=127.0.0.1
DB_PORT=5432
DB_DATABASE=app1
DB_USERNAME=postgres
DB_PASSWORD=123456789
```

Run Laravel



- Run migration to test the connection:

```
php artisan migrate
```

- Run the server:

```
php artisan serve
```

- Test it in browser:

<http://127.0.0.1:8000>

